

# Power Reliability

## 1. Overview

The lightning detection unit is powered by solar with a battery reserve, and it sends its data to the secure cloud service over the internet using its own built-in communications hardware. The unit does not connect to, or require any access to, the client's WiFi network, LAN, or IT infrastructure.

## 2. Power Source and Reserve

Energy is supplied by a solar panel through intelligent power management circuitry that charges the battery pack, protects it, and provides a stable supply to the unit. The battery reserve carries the unit through the night and through periods of poor sunlight.

| Item                   | Detail                                                                                                            |
|------------------------|-------------------------------------------------------------------------------------------------------------------|
| Primary energy source  | Solar panel (10 W recommended minimum)                                                                            |
| Energy management      | Maximum Power Point Tracking (MPPT) charging, automatic charge/discharge change-over, and full battery protection |
| Battery reserve        | 3 x 18650 lithium-ion cells, 3000 mAh each, in parallel                                                           |
| Total reserve capacity | 9 000 mAh at 3.7 V (about 33 Wh; 25 Wh usable after protection limits)                                            |

## 3. Battery Backup Duration

The figures below cover the complete self-contained unit, including its detection electronics and its internal cloud-communications hardware, all running from the battery reserve with no solar input at all (e.g. night-time plus fully overcast conditions).

|                                                    | Estimated backup (no sunlight) |
|----------------------------------------------------|--------------------------------|
| Complete unit, normal operation                    | Approximately 10 to 12 hours   |
| Complete unit, communications in low-power standby | Up to approximately 1.5 days   |

## 4. Built-in Power Resilience

The unit restarts and re-initialises itself after any power loss, including full battery depletion

On-board storage is checked and repaired automatically after sudden power loss, preventing data corruption.

Deep-discharge protection: the power circuitry protects the battery from harmful over-discharge and over-charge.

The unit reports its status and temperature hourly (visible on the admin panel)